COURSE MATERIAL

INTERNET OF THINGS

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CH1: THINGS & CONNECTIONS

1.1: what is things

There are many different types of IoT-enabled devices available. However, most IoT devices use sensors, controllers, and actuators to perform functions.

Cisco IoT Solutions
Cisco IoT System reduces the complexities of digitization.
Six Pillars of the Cisco IoT System are:

1.Network Connectivity
2.Fog Computing
3.Cybersecurity and Physical Security
4.Data Analytics
5.Management and Automation
6.Application Enablement Platform

Overview of a Controlled System

Feedback loops	used to provide real-time information to its controller based on current behavior.
In a closed loop	feedback is continuously being received by the controller from its sensors.
The controller	continuously analyzes and processes information, and use actuators to modify conditions.

Sensors

- A sensor is a device that can be used to detect some type of information to sense information from the real world environment.
- A sensor may be connected to a controller either directly or remotely.
- Sensors send data to a controller. That controller could react to the sensor data immediately and change the actuator settings.

Actuators

- An action/actuator is a basic motor that can be used to move or control a mechanism or system, based on a specific set of instructions.
- There are three types of actuators:

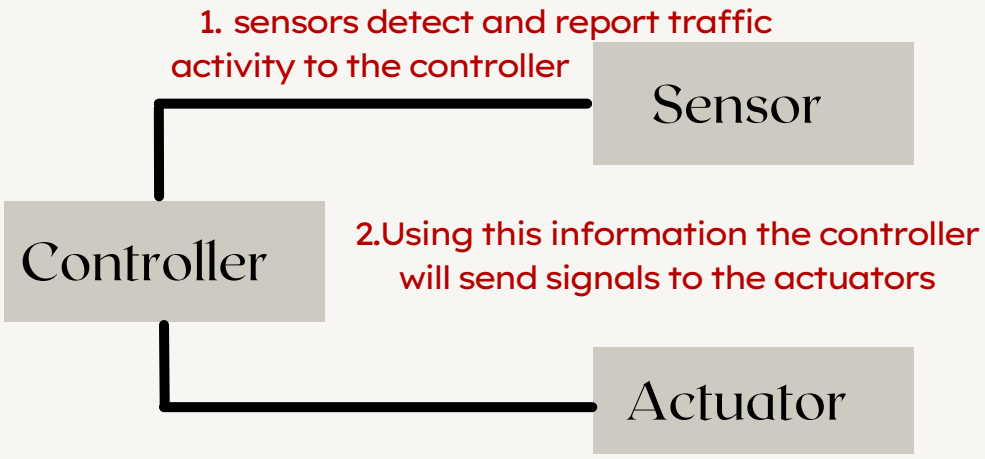
1. Electrical - Powered by a motor that converts electrical energy into mechanical operations
2. Hydraulic - Uses fluid pressure to perform mechanical movement
3. Pneumatic - Uses compressed air to enable mechanical operations

Ex of it: camera, sirens, alarm

Controllers

- Responsible for collecting data / instructions from sensors and providing network connectivity.
- Controllers may have the ability to make immediate decisions.
- May also send data to remote and more powerful computer for analysis.
- It's save Data in cloud for future analysis

sample IoT topology / smart traffic lights



Controller Types

Arduino microcontroller	Raspberry Pi (RaPi)
• Operate without internet , uses hobbyists and professionals. • Requires Less Power • More suitable for analog input	• Operate without internet , uses hobbyists and professionals. • Requires More Power • Less suitable for analog input
The key difference between the two is physical size, available processing power, memory, and OS	

IoT Process Flow

- A simple IoT system include sensors connecting, through a wireless or wired connection, to actuators or controllers.
- Some devices can have more than one function.

Device or Controller	Edge Node	Fog Node	Data Center	Cloud
Generating data	Capturing data	Aggregate data	Leveraging data	Analyzing data

Questions:

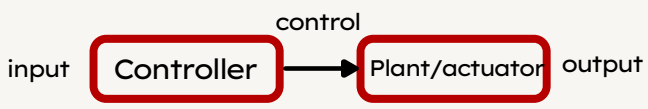
DEVICE	DESCRIPTION
Controller	Adds intelligence to things by collecting data and either making immediate decisions based on data collected or passes data to a more powerful computer for analysis
Actuators	A basic motor that can be used to move or control a mechanism or system
Arduino	Commonly used to create IoT projects by home hobbyist and is suitable in situations where vou want to connect simple sensors a read real time data
Sensors	Can be used to measure a physical property and detect some tvpe of information from the physical world
Raspberry Pi	A more powerful controller often used by hobbyists and suitable to use when a large amount of data needs to be processed

Controlling system

Open Loop

Ex: switch on the light and the light opens and finish

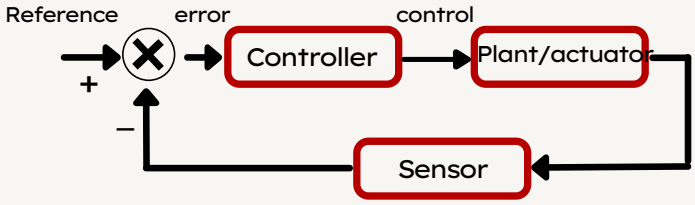
- do not use feedback
- Instructions and action
- Controller & actuator



Close Loop

Ex: switch on the light and the light opens and will check every time and gives feedback again and again

- uses feedback
- Controller, actuator & sensor



1.2: what are connections

Models of Communication

Layered networking models are used to illustrate how a network operates

Advantages

- Assist in protocol design because protocols that operate specific layer have defined information that they act upon and a defined interface to the layers above and below.
- They foster competition because products from different vendors can work together.
- They prevent technology or capability changes in one layer from affecting other layers above and below.
- They provide a common language to describe networking functions and capabilities.

TCP/IP MODEL	DESCRIPTION
Application	Represent data to the user, plus encoding and dialog control
Transport	supports communication between various devices across diverse networks
Internet	Determine the best path through the network
Network Access	Controls the hardeware device and media that make up the network

OSI MODEL	DESCRIPTION
Application	contains protocols used for process-to-process communication
Presentation	give for common representation data transfer between application layer Services
Session	gives Service to presentation to organize dialogue, manage data exchange
Transport	defines service to segment, transfer, reassemble data for individual communication between end devices
Network	gives Services to exchange pieces fo data over network between identified end devices
Data Link	Describe Methods for exchange data frame between devices over common media
Physical	Describe mechanical, electrical, functional, procedural means to actives , maintain , de-active physical connections for bit transmission to and from a network device

1. Network connections are most often described as using the OSI and TCP/IP models. As shown in Figure 1, both models use layers, and each layer has a function.
2. The TCP/IP protocol model The architecture of the TCP/IP protocol suite follows the structure of this model which is why it is called a protocol model. The TCP/IP protocol model is commonly referred to as the Internet model.
3. The OSI reference model provides an extensive list of functions and services that can occur at each layer.also describes the interaction of each layer with the layers directly above and below.
4. both models are often used interchangeably.
5. TCP/IP model layers are referred to only by name, the seven OSI model layers are more often referred to by number rather than by name.

IoT World Forum Reference Model

the parts are called levels instead of layers The IoT reference model was developed as a common framework to guide and to help accelerate IoT deployments. The model is the result of a collaborative effort of the 28 members of the IoT World Forum's Architecture, Management and Analytics Working Group. It is also endorsed by the Industrial Internet Consortium (IIC).

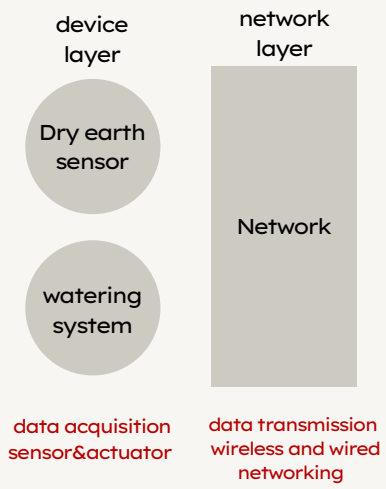
The intent of the IoT reference model is to provide common terminology and help clarify how information flows and is processed for a unified IoT industry.

LEVEL		DESCRIPTION
7	Collaboration & Processes (Involving people and business processes)	Transcends multiple applications to include the communication and collaboration required between people and business processes.
6	Application (Reporting, analytics, control)	Information interpretation based on the nature of the device data and business needs.
5	Data Abstraction (Aggregation and access)	Focused on rendering the data and its storage in ways to enable application development.
4	Data Accumulation (Storage)	Data in motion is converted to data at rest. The data is also transformed so that it can be consumed by upper levels.
3	Edge (Fog) Computing (Data element analysis and transformation)	Converts data into information that is suitable for storage and higher level processing.
2	Connectivity (Communication and processing units)	Responsible for reliable and timely data transmission between devices and the network, across networks, and between the network and data processing in Level 3.
1	Physical Devices & Controllers (The "Things" of IoT)	Includes a wide range of endpoint devices that send and receive information.

Simplified IoT Architecture

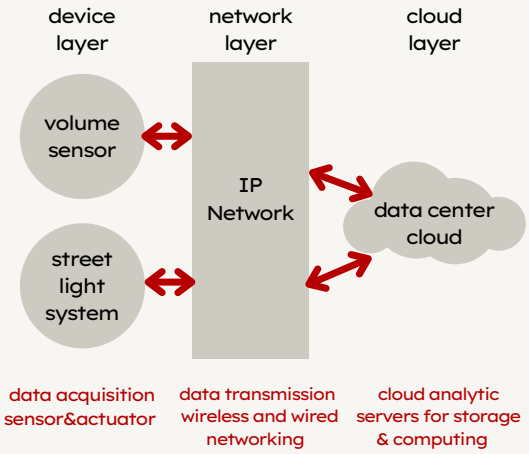
- Device-to-Device

IoT solutions often support one smart object connecting directly to another via a wireless protocol such as Bluetooth or Zigbee. An example of this level is a sensor that is located in a vineyard and detects dry soil. It sends a signal to an actuator that triggers the watering system.



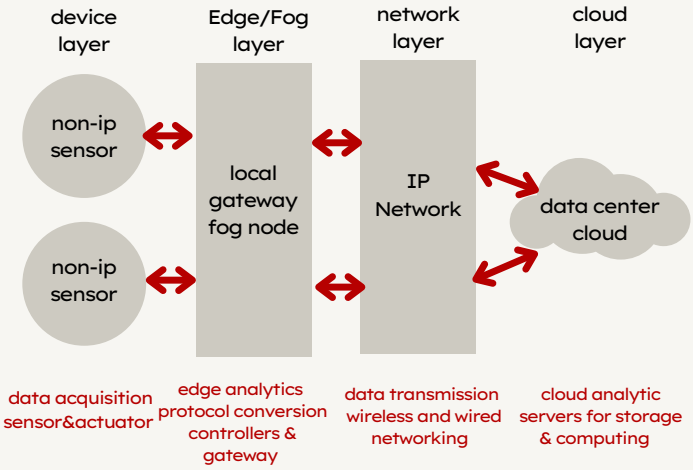
- Device-to-Cloud

In a device-to-network-to-cloud communication model, the IoT device connects through a local network directly to an Internet cloud service using traditional wired Ethernet or Wi-Fi connections. This model establishes a connection between the device, the IP network, and the cloud to allow the exchange of data and control messages.



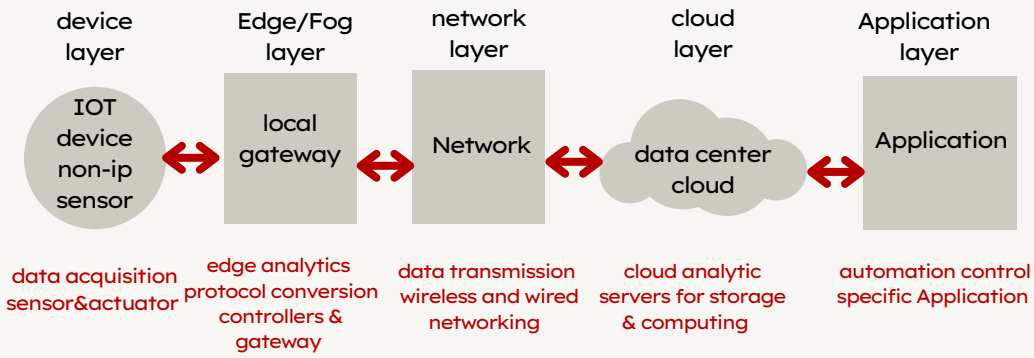
- Device-to-Gateway-to-Cloud

Many smart devices, such as fitness trackers, are not IP-enabled and do not have the native ability to connect directly to the fog or the cloud. For these devices, there is application software operating on a local gateway device which acts as an intermediary between the device and the cloud service. The gateway may also provide security and data or protocol translation. For devices, like fitness trackers, the gateway is often an application running on a smartphone



- Device-to-Gateway-to-Cloud-to-Application

Another connection option supports smart device data collection and transfer through a gateway to a local IP network. The data then flows to the fog or the cloud and is then available for users to export and analyze. The data is often analyzed in combination with data from other sources or other cloud services.



Knowledge of the four basic levels of connections will ensure that consideration is given to device interoperability and open standards. These are key considerations when developing an internetworked IoT system.

Raspberry Pi connectied to breadboard

Connections Within Networks

- Connections can have different contexts.
- Power connections, circuit connections or network connections.

Physical Connections

•Relate to the media and cable type.

•Common media types include copper, fiber optics and wireless.

Physical Connections

There are three types of media used by IoT devices to communicate:

Copper -

- Networks use copper media because it is inexpensive, easy to install, and has low resistance to electrical current.
- Copper media is limited by distance and signal interference. Therefore, all copper media must follow strict distance limitations as specified by the guiding standards.

Fiber optics –

- Fiber-optic cables can travel significantly longer distances than copper cables due to their immunity to signal interference. The sending device transmits the binary bits as light pulses using LEDs or lasers. The receiving device uses photodiodes to detect the light pulses and convert them to voltages. Fiber-optic cables are broadly classified as single-mode fiber (SMF) and multimode fiber (MMF). SMF consists of a very small core using expensive laser technology to send a single ray of light over long distances spanning hundreds of kilometers. MMF consists of a larger core and uses lower-cost LED emitters to send light pulses. MMF is popular in LAN deployments because it can support 10 Gb/s over 550 meters links.

Wireless –

- Wireless consists of a wide range of connection options including electromagnetic signals, radio and microwave frequencies, and satellite links.

All copper and fiber cable connectors adhere to specific physical layer standards. These standards specify the mechanical dimensions of the connectors and the acceptable electrical properties of each type. These network media also commonly use modular jacks and plugs to provide easy connection and disconnection.

Data Link and Network Connections

Network communication requires protocols to establish the rules of communications. Data Link protocols:

- Allow the upper layers to access the media
- Prepare network data for the physical network
- Control how data is placed and received on the media
- Exchange frames between nodes over a physical network media, such as copper or fiber-optic
- Receive and directing packets to an upper layer protocol
- Perform error detection

The most popular data link layer connection used in wired networks? is Ethernet.
Other data link protocols include wireless standards such as IEEE 802.11 (Wi-Fi), IEEE 802.15 (Bluetooth), and cellular 3G or 4G networks.
LoRaWAN and NB-IoT are examples of emerging IoT supporting technologies.

Application Connections

- The IoT supports many types of connections.
- Devices must use the same application layer protocols to connect.
- The application will vary depending on the devices and type of connection involved.
- MQTT and REST are newer application protocols, created to support IoT devices that connect in the myriad of different types of remote configurations.
- MQTT is a lightweight messaging protocol with minimal overhead that provides high data integrity and security for remote environments.
- REST or RESTful web services is a type of API designed to make it easier for programs to interact over the Internet.

What is Metadata?

- Metadata refers to the data about data.
- Metadata can be embedded within a digital object or it can be stored separately.
- Metadata is not usually seen by a user.

The Impact of IoT on Privacy
Suggestions and design considerations concerning privacy include:

- Transparency
- Data Collection and Use
- Data Access

Challenges for Securing IoT Devices
Some IoT network security impacting factors include:

- Increasing Number of Devices
- Non-Traditional Location of Devices
- Changing Type and Quantity of Gathered Data
- Lack of Upgradeability

CH 2 : SENSORS, ACTUATORS, AND MICROCONTROLLERS

- What is Electronics?

Electronics is the field of study focused on the control of electricity and the physical components and circuits that help direct electrical energy

- Definitions/ Terms commonly used in electronics include
  1. Electrons, atoms, and chemical elements
  2. Electric current
  3. Electrical conductors, insulators, and circuits
  4. Voltage, Amperes (amps), and Power

	DESCRIPTION
Electric current	is created from the movement of electrons. Current flows in a closed loop and is constant everywhere in that loop.
Electrons	along with protons and neutrons are what make up atoms. The basic charge on an electron is measured in terms of coulombs. One coulomb of charge is equal to the amount of charge carried by one ampere in one second
Atoms	are the building blocks of all elements and matter. Electrons carry negative charges and are attracted to the positively charged protons within the nucleus of the atom.
Chemical elements	on the periodic table are made up of different types of atoms. The attraction between atoms and their outer electrons is stronger in some elements than in others.
Electrical conductors	are materials with elements that have a weak attraction between atoms and their electrons. In conductive elements, electrons tend to move from atom to atom. Examples of electrically conductive materials are metals like copper, gold, and silver
Electrical insulators	are materials made up of elements that strongly attract their electrons and in which the electrons never leave the atom. Examples of materials that are electrical insulators are dried wood, glass, and various rubber materials.

Voltage	is the force that drives current. It can also be referred to as electric pressure. Voltage is measured as the difference in electric potential energy between two points.
Amperes (Amps)	measure the strength of electric current. Amperes are a measure of the number of electrons in an electric current. One ampere per second is equivalent to one coulomb traveling through a circuit in one second. This is roughly equivalent to the flow of 6.241509×10^18 electrons per second.
Power	is the amount of energy consumed over time. Power is measured in Watts. The basic formulation of power is power = voltage x current.
electrical circuit	is a physical network (or model of a physical network) of interconnected electrical components including batteries, resistors, capacitors, inductors, and switches

- Ohm's Law

1. Ohm's Law states that within a circuit, voltage (V) is directly proportional to the strength of current (I) multiplied by resistance (R).
2. Resistance is measured in ohms (Ω) omega

- Basic Circuit

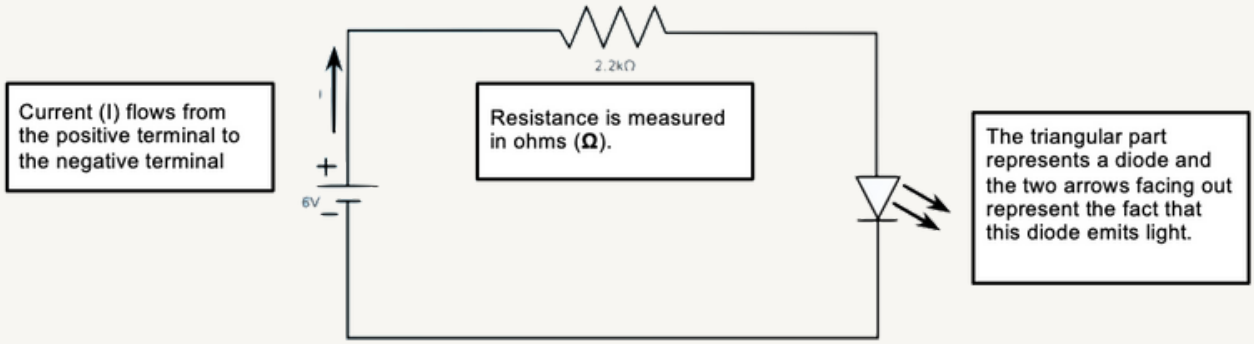
1. An electrical circuit is a closed conductive path that allows electrons to flow and create an electric current.
2. A circuit also needs an electrical energy source like a battery to start the flow of electricity.

$$\boxed{V = I * R} \quad \boxed{R = V/I} \quad \boxed{I = V/R}$$

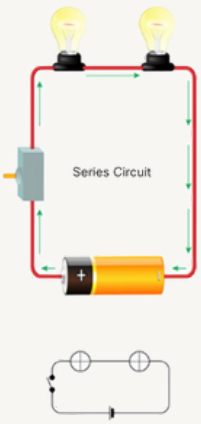
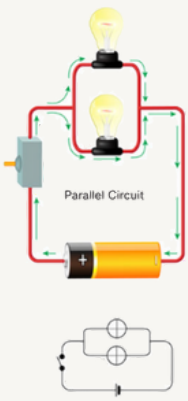
V - Voltage - Force that drives the current
R - Resistor - limits the floor of power spliy

- The following circuit diagram (schematic) consists of:

1. 6 volt (V) battery provides current
2. 2.2 kΩ resistor (protects the LED from receiving too much current and being destroyed)
3. A light-emitting diode (LED)



- Series and Parallel Circuits

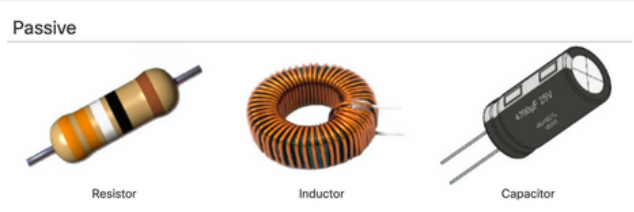
Series Circuit:	Parallel Circuit:
components are interconnected one after another in a path between the positive and negative terminals of the power source connected one after other if one lost will not work	current flows from the battery terminals but splits at a junction which leads to parallel pathways through the circuit. Component connected along each pathway each get their own share of current
	
The electric current travels through each component in a linear fashion	each component gets its own supply of current, if any one component or LED were to fail, it would not stop the current from flowing to the other pathways and powering the other components or LEDs

Passive, Active, Linear, and Nonlinear Circuits

- Active circuits contain active components; components that rely on external power source to control current flow.
- Passive circuits contain passive components; components incapable of controlling current flow.
- It just store it.
- Analog circuits are circuits where the signal is contiguous.

Direct Current vs. Alternating Current

- In DC current, electron flow is only in one direction.
- Batteries, power supplies, thermocouples, solar cells, or dynamos generate DC.
- In AC current, electron flow periodically reverses direction.
- Hydroelectric plants generate AC.



Analog Circuits vs. Digital Circuits

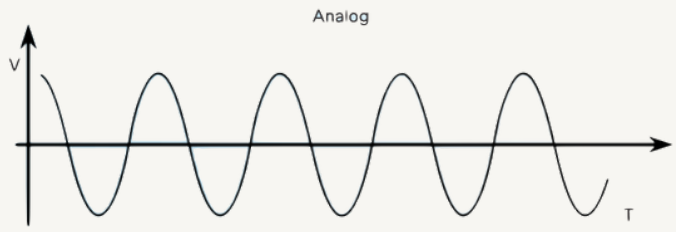
1. Analog Circuits: Circuits in which signals vary continuously with time.
2. Digital circuits: Circuits in which signals that take one of two discrete values. (ON/OFF or 1, 0)

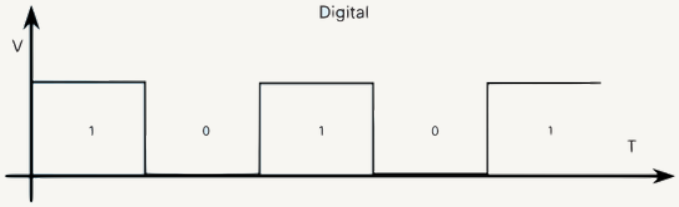
Components

1. Electronic components are specialized devices used in a circuit to control current.
2. Components have two or more electrical terminals (leads) that enable them to connect to an electronic circuit.

Larger Electronic Building Blocks

1. Solenoids can be used to electrically open door latches, open or shut valves, move robotic limbs, and even actuate electric switch mechanisms.
2. Relays allow for controlling a large amount of current and/or voltage with a small electrical signal.





Design Phase:

- Consists of three steps: Concept, Research, Circuit Design.
- A circuit diagram shows the components and interconnections of the circuit using standardized symbolic representations.

Prototype Phase:

- Consists of four steps: Hardware, Mechanical, and Software Development, PCB layout, Build prototypes, Product Testing
- A solderless breadboard is a tool commonly used in electronic prototyping.

Production Phase:

- Consists of three steps: Production Readiness Review, Production, On-going Maintenance.
- Often employ on printed circuit boards (PCBs).

Introduction to the SparkFun Inventor's Kit (SIK)

This is a starter kit for building circuits and includes:

- Solderless breadboard
- SparkFun RedBoard (Arduino-like board)
- Various resistors, diodes, LEDs, sensors and actuators
- Connecting wires (jumper wires, mini-B cable, ...)

Arduino Microcontroller

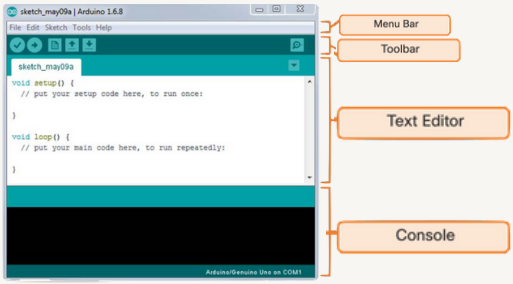
- The Arduino is a popular microcontroller for prototyping.
- Instructions for the Arduino are programmed using the Arduino integrated development environment (IDE).
- The SparkFun RedBoard is an Arduino-like board that can be programmed using Arduino IDE.

Building a Circuit

- A simple circuit can be created by:
- Connecting electronic components (LED, resistor, and jumper wires) in series along a row on the breadboard.
- Connecting the power source to the lower red and black jumper wires.
- This should complete the circuit and light the LED.

The Arduino IDE

- Free, downloadable software used to interact with the Arduino board.



Menu bar : open file, save
Tool bar : compile and verify programs upload
Text Editor : type program, copy,paste,replay

Console : see error msg and output

Writing code

- Programs written using the Arduino IDE are called sketches and are saved with the file extension of .ino.
- Arduino sketch keywords can be divided in three main category types: structures, values (variables and constants), and functions.
- Keywords used include void, setup(), loop() function, and more.

Testing

- To test and verify the sketch code, click on the checkmark toolbar icon.
- The IDE compiles the code and checks for syntax errors.
- To upload the sketch to the Arduino and test the code, click on the second toolbar icon (->)

Sensors

- Devices that detect an event from the physical environment and respond with electrical or optical signals as output.
- The SIK contains various sensors including Soft potentiometer, Flex sensor, Photo resistor and Temperature sensor.

Actuators and Relays

- An actuator is a type of motor that is responsible for creating movement.
- The SIK includes two types of electric actuators that convert electrical energy into mechanical torque.
- A relay is an electrically controlled mechanical switch.
- The SIK includes a plastic box that contains an electromagnet that causes a switch to trip when it receives a current.

How Everything Connects in PT

- Packet Tracer 7.x can be used as a prototyping tool.
- There is a new group icon contained in Packet Tracer version 7.1 that is labeled Components.
- The PT IoT boards contains an MCU and a SBC.
- The MCU and SBC are similar to an Arduino and a Raspberry Pi, respectively.
- There are also actuators and sensors that can be used in prototypes.
- The IoE Custom Cable found in the Connections group can be used to connect IoT things to an MCU board.

CH 3 : PROGRAMMING

What is a Program

- Code is a set of ordered instructions created to accomplish a specific task.
- A bread recipe can be seen as a program.
- Computer programs can be written in different programming languages.

Ex of Operation System. : Windows, Linux, Unix, Android , IOS,...

Programs are Everywhere

- All computers need programs.
- Operating Systems, firmware, and applications are examples of programs

Why Learn Code?

- Programmers are valued in the job market.
- Today, programmers may work on firmware, device
- drivers, mobile applications, web interfaces, data analysis, and more.
- Programmers can create their own tools.

interpreter and compiler (there are translator)

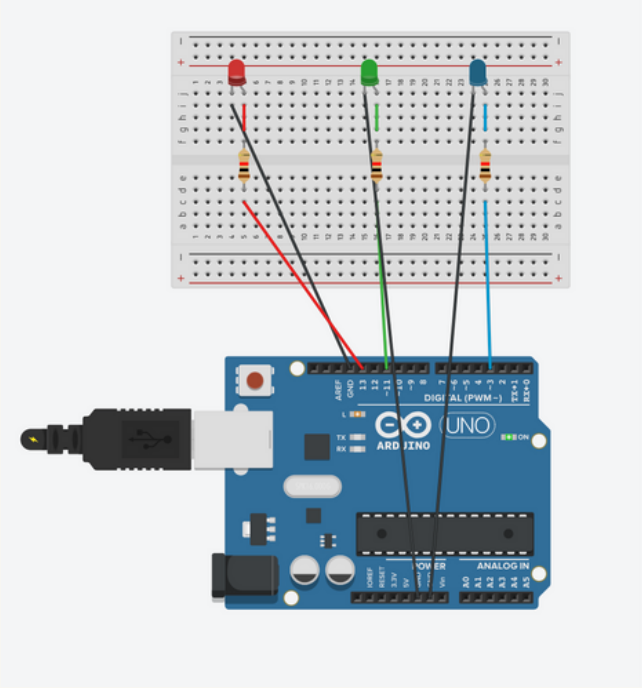
interpreter:	compiler:
The execution of the program takes line by line (slower then compiler)	convert the instructions The execution of the program happens only after the entire program is compiled as a group (faster)
programming languages. PHP , JavaScript, python easy for troubleshooting	Programming languages C, C++, Java

Computer Languages

- There are several different computer languages.
- Some computer languages are better than others at certain types of tasks.
- computer languages.
  JavaScript, Python, Blockly, C, and Java

Linux- file

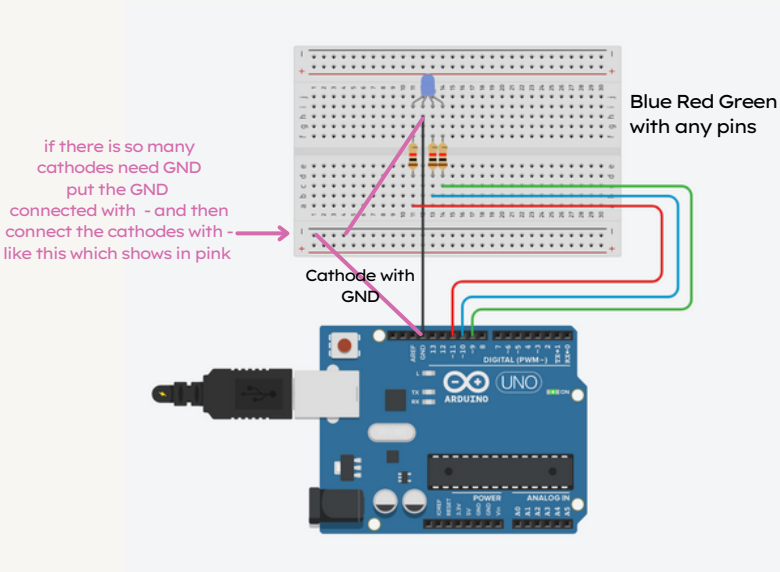
Arduino- Sketch-> . IDE name of the software but we save it as a program .INO



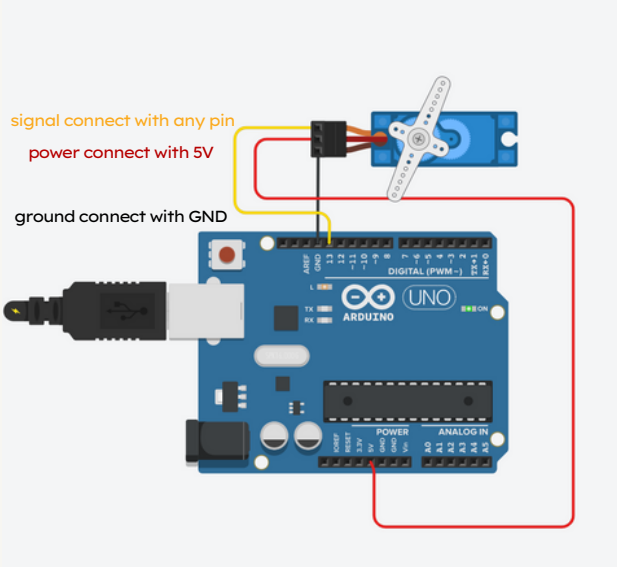
```
// C++ code
//
void setup()
{
  pinMode(11, OUTPUT);
   pinMode(13, OUTPUT);
   pinMode(3, OUTPUT);
}

void loop()
{
  digitalWrite(11, HIGH);
  delay(100); // Wait for 1000 millisecond(s)
  digitalWrite(11, LOW);
  delay(100); // Wait for 1000 millisecond(s)

  digitalWrite(13, HIGH);
  delay(100); // Wait for 1000 millisecond(s)
  digitalWrite(13, LOW);
  delay(100); // Wait for 1000 millisecond(s)

  digitalWrite(3, HIGH);
  delay(100); // Wait for 1000 millisecond(s)
  digitalWrite(3, LOW);
  delay(100); // Wait for 1000 millisecond(s)
}
```

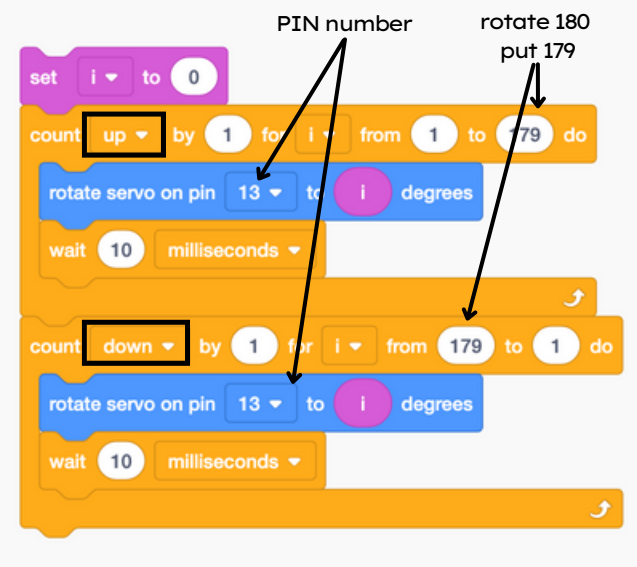
Cathode connected with GND and Anode with resistor then any pin

digitalWrite , analogWrite : perform an action
digitalRead , analogRead : to read (sensor)




```
// C++ code
//
void setup()
{
  pinMode(9, OUTPUT);
  pinMode(10, OUTPUT);
  pinMode(11, OUTPUT);
}

void loop()
{
  digitalWrite(9, HIGH);
  delay(100); // Wait for 1000 millisecond(s)
  digitalWrite(9, LOW);
  delay(100); // Wait for 1000 millisecond(s)

  digitalWrite(10, HIGH);
  delay(100); // Wait for 1000 millisecond(s)
  digitalWrite(10, LOW);
  delay(100); // Wait for 1000 millisecond(s)

  digitalWrite(11, HIGH);
  delay(100); // Wait for 1000 millisecond(s)
  digitalWrite(11, LOW);
  delay(100); // Wait for 1000 millisecond(s)
}
```

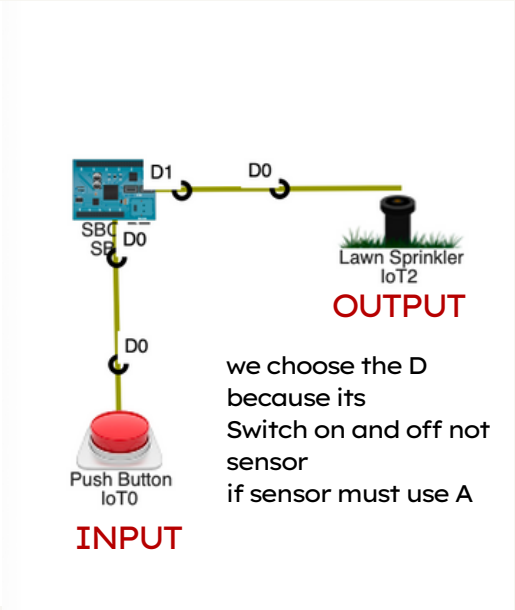




Packet Tracer

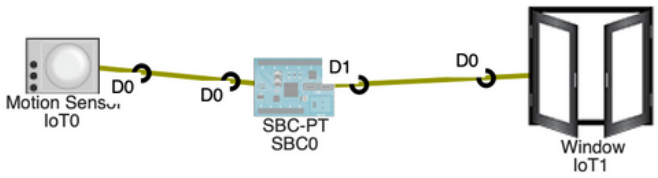
when you push the button the water will spread

```python
from gpio import *
from time import *

def main():

    pinMode(1, OUTPUT)
    pinMode(0, INPUT)

    while True:
        value = digitalRead(0)
        if(value>0):
            customWrite(1, 1)
        else:
            customWrite(1, 0)

if __name__ == "__main__":
    main()
```

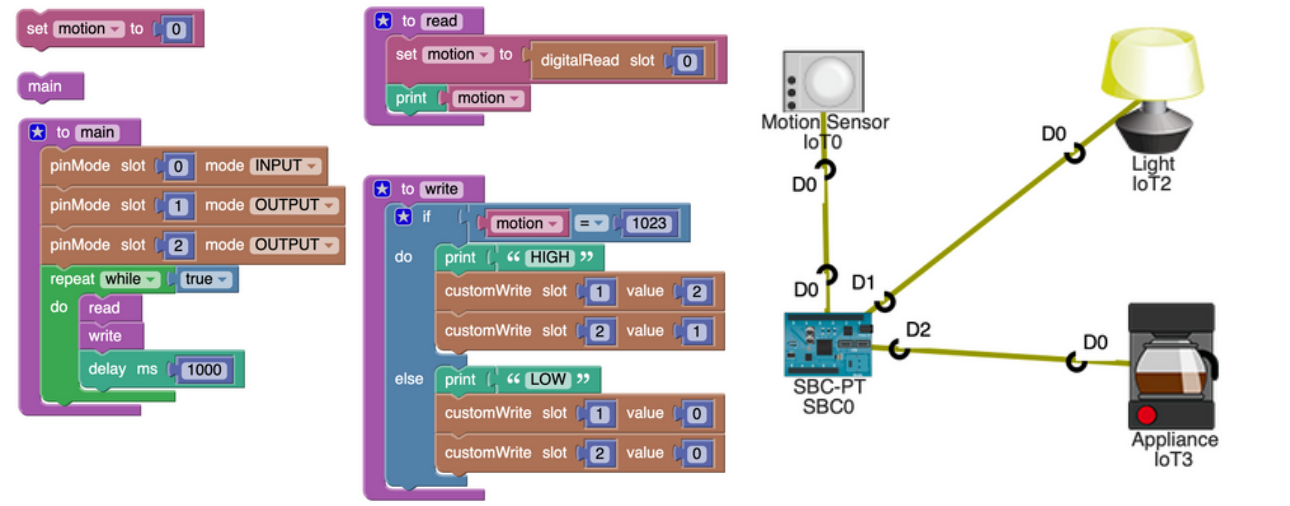


when you alt move in sensor the window will open

```python
from gpio import *
from time import *

def main():
    pinMode(0, INPUT)
    pinMode(1, OUTPUT)

    while True:
        d=digitalRead(0)
        print(d)
        if(d>0):
            customWrite(1,1)
        else:
            customWrite(1,0)

if __name__ == "__main__":
    main()
```



when you alt move in sensor the lamp will open and coffee machine



The Raspberry Pi and its Ports

- The Pi is a small and inexpensive computer.
- It has a number of USB ports that can be used to connect various devices including keyboards, mice, external drives and cameras.
- The Pi includes an 10/100Mbps Ethernet port and 40 GPIO pins, operating at 3.3V.
- Other Pi ports include an audio out, a micro SD card slot, and a micro USB (used for power) connector.

The Pi3 also adds:

- 1.2 Ghz 64-bit quad-core ARMv8 CPU
- 802.11n Wireless LAN
- Bluetooth 4.1
- Bluetooth Low Energy (BLF)

The Pi can run a number of operating systems, including Linux and Windows.

s1 : install OS image on the micro SD card
s2: place SD slot of the Rapi
s3: connect a USB
s4: connect a monitor or tv using HDML port
s5: power device with power adapter
s6: install app called PL (prototype lab)App

Linux

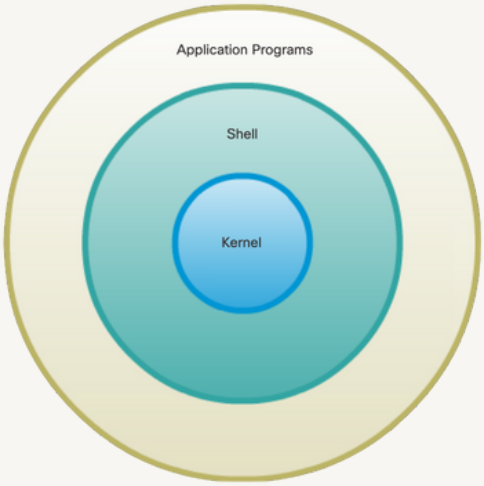
Understanding Linux

- Linux is open source, fast, reliable and small and requires very little hardware resources to run.
- Linux is part of several platforms; from wristwatches to supercomputers.
- Linux distributions include the Linux kernel, plus a number of customized tools and software packages.
- Debian, Red Hat, Ubuntu and Slackware are just a few examples of Linux distributions.
- Raspbian is a Linux distribution based on Debian and created specifically for the Raspberry Pi.

Accessing the Linux Shell

- The Linux operating system can be divided into kernel and shell.
- The shell is a command interpreter.
- The shell is text based and also called CLI (command line interface)

even thoughwe have GUIs



kernel : main part (core of OS)
shell : write the program(that run)
app programs:

CLI : Bourne Shell(sh), Bash(bash), C Shell(csh) , improve CShell (tcsh), Z Shell(zsh) are popular Shells

GUI: Terminator , term, xterm , console , gnome-terminal

File Permission :

- most everything is treated as a file
- file permission provide a mechanism to define permission to file
- possible permission right are Read ,write and execute and can be defined for the user who own the file the group and other system users
- The Root User can Override file Permission

Binary	Octal	Permission	Description
000	0	---	No access
001	1	--x	Execution only
010	2	-w-	Write only
011	3	-wx	Write and Execution
100	4	r--	Read only
101	5	r-x	Read and Execution
110	6	rw-	Read and Write
111	7	rwx	Read, Write and Execution

ifconfig	Used to display or configure network card related information. If issued without parameters, **ifconfig** will display the current network card(s) configuration.
apt-get	Used to install, configure and remove packages on Debian and its derivatives. Note: **apt-get** is a user-friendly command line front-end for **dpkg**, Debian's package manager. The combo **dpkg** and **apt-get** is the default package manager system in all Debian Linux derivatives, including Raspbian.
iwconfig	Used to display or configure wireless network card related information. Similar to **ifconfig**, **iwconfig** will display wireless information when issued without parameters.

Creating & Viewing Files	**1. cat > filename** **2. Add content** **3. Press 'ctrl + d' to return to command prompt.**
Viewing content of the file	**Cat filename**
Listing files	**ls**
Deleting Files	**rm filename**
Moving files	**mv filename newfilename**
Copy file	**cp firstfilename secondfilename**
Creating Directories	**mkdir**
Removing Directories	**rmdir**
Renaming directories.	**mv directoryname newdirectoryname**
manual which is a reference book of a Linux operating system. It is similar to HELP file found in popular software. To get help on any command that you do not understand, you can type man The terminal would open the manual page for that command.	**Man**
The clear command	**clear**
shutdown	**reboot**
For Search Output	**grep**
Change the Password	**passwd**
Current directory	pwd
Modify the permission of the file	**Chmod**
Change the ownership of the file	**chown**

summary

Syntax
void setup {}
void loop {}
...

Tinkercad
For Bulb RED - POSITIVE , BLACK - NEGATIVE
if we put the LED with the Battrey will broke it have to be connected with a resistor

buliding blocks
sensor , actuator, controller

visible deference's of
Adriano(micro mcu) digital and analog (1 usb) (OS : IDE)
Raspberry Pi pi(single sbc) 40 pins (digital only) (4 usb) support wireless, bluetooth (SD card)

There is a component which can read the temperature of the room: sensor
There is a component which make noise when temperatures increase : actuator
here is a component which you give instructions : controller

senior :
in the collage we have labs and class suddenly there is fire which actuator ans sensor you use : fire detector sensor and actuator: alarm (to modify the ppl)or water sprinkler (to reduce the fire)

seniro :
auto garden : water level sensor to check the water
and humidity to check the soil
Actuator : if humidity is high > use water pump or alerm

open loop > have controller and actuator (off and on) no feedback
close loop > have sensor, controller and actuator (feedback)

Positive feedback > keep on checking till achieve the target
Negative feedback > will happend one time then will be different

use of MQTT --> light weight msg protocol used to transfer msg between the iot devices(low band width , power, transferred devices)

Rest : type of API used to make easy and interact
mitadata :

Challenges ..
 ch 2 summary
what is atom > block of matter
3 sub of atom 1. electron(negative) 2 .proton(positive) 3.neutron
when electron move the electricity is produced

seriale circuit --> component connected one after another
parallel circuit --> current flow on all path

active-> take the current and perform action (diode, Transistor, SCR)
passive -> only store the current and give (Resistor , Inductor, Capacitor)

Circuts Analog -> current voltage may continusouly (sensor , amplifier)
Circut Digital -> take discrite value (on / off)

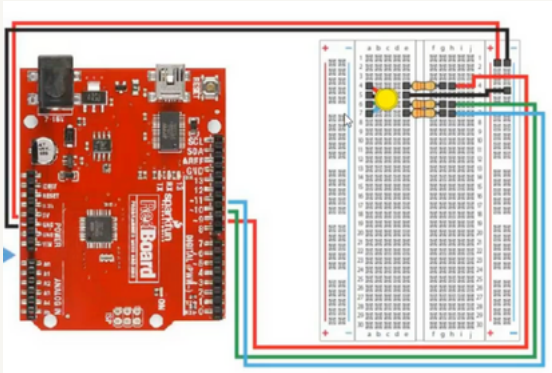
design
1. digrams
2. prototype (using by PCB printed circuit board)
3. programing

solderless breadboard -> plug in and out and arrange row and coloums useful

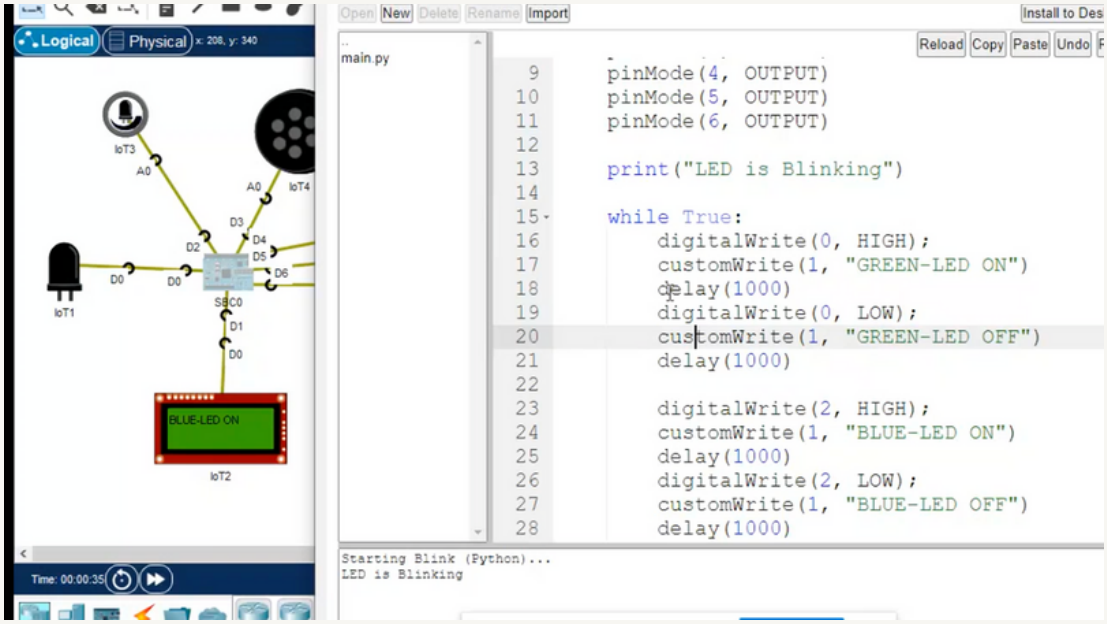
The Arduino IDE App
4 areas - Menu bar , Toolbar, TextEdit , Console

purpose of red and black

for power supply



program > set of ordered instructions to accomplish a task
difference interpreter and compiler


Logical
Physical
x: 208, y: 340
IoT3
A0
A0
IoT4
D3
D2
D4
D5
D6
D0
D0
IoT1
SSC0
D1
D0
BLUE-LED ON
IoT2
Time: 00:00:35
Open
New
Delete
Rename
Import
Install to Des
Reload
Copy
Paste
Undo
main.py
pinMode(4, OUTPUT)
pinMode(5, OUTPUT)
pinMode(6, OUTPUT)
print("LED is Blinking")
while True:
digitalWrite(0, HIGH);
customWrite(1, "GREEN-LED ON")
delay(1000)
digitalWrite(0, LOW);
customWrite(1, "GREEN-LED OFF")
delay(1000)
digitalWrite(2, HIGH);
customWrite(1, "BLUE-LED ON")
delay(1000)
digitalWrite(2, LOW);
customWrite(1, "BLUE-LED OFF")
delay(1000)
Starting Blink (Python)...
LED is Blinking

CH 4 : NETWORK , FOG AND CLOUD COMPUTING

local-area networks (LANs) : geographical distance is less
wide-area networks (WANs). : geographical distance is More

End Devices : Desktop computer, Laptop, Printer, IP Phone, Wireless Tablet, telepresences Endpoint

LAN and WAN

- The path from source to destination can be a single cable or a collection of networks.
- A Personal Area Network (PAN) is a type of network that usually spans a few meters around an individual and is often used in IoT.
- A Local Area Network (LAN) is a type of network infrastructure that spans a small geographical area and is used to connect end devices..
- A LAN is normally a high-speed network under the control of a single administrative entity.
- A Wide Area Network (WAN) is a type of network infrastructure that spans a wide geographical area and is used to connect WANs.
- A WAN is normally a low-speed network and may include portions from different Internet Service Providers (ISPs)
- LANs often connect machines in the factory plant.
- WAN devices have evolved to create Low PowerWide Area Networks (LPWAN) for use in the IOT

network components:

- Devices
- Media
- Services

Network Devices and Communication Media

- Network devices are devices that connect to each other through a network.
- An end device is either the source or destination of a message transmitted over the network.
- Intermediary devices connect the individual end devices to the network and can connect multiple individual networks to form an internetwork.
- Network addresses are used to uniquely identify devices on a network.
- Network media provide the physical channel over which the message travels from source to destination.
- Network media Type

  - Metallic wires within cables - data is encoded into electrical impulses
  - Glass or plastic fibers (fiber-optic cable) - data is encoded as pulses of light
  - Wireless transmission - data is encoded using radio waves

Network Protocols

- Devices must conform to common protocols before they can communicate.
- Two very important network protocols are Ethernet and IP.
- Ethernet rules enable communication between local devices.
- IP enable communication between remote devices.

Basic Routing

- Network packets must often transverse several networks to get to the destination.
- Routing is the process of directing a network packet to its destination.
- Routers are intermediary network devices that perform routing

LANs, WANs and the Internet

- Single router designs are common in SOHO.
- The single router connects SOHO devices to the Internet.
- The single router is the default gateway for all SOHO devices.

IoT Protocols

- IoT Devices are often embedded devices designed to work in sub-optimal conditions.
- These devices require specialized protocols to function with low power and limited connectivity.
- IoT devices use CoAP (Constrained Application Protocol) and MQTT (Message Queuing Telemetry Transport).

CoAP is based on HTTP and the REST model where resources are retrieved from a server using URIs/URLs, The clients use the well-known methods of GET, PUT, POST, and DELETE to manipulate these resources. CoAP can be used via other mechanisms, such as SMS on mobile communication networks.

Securing the Network

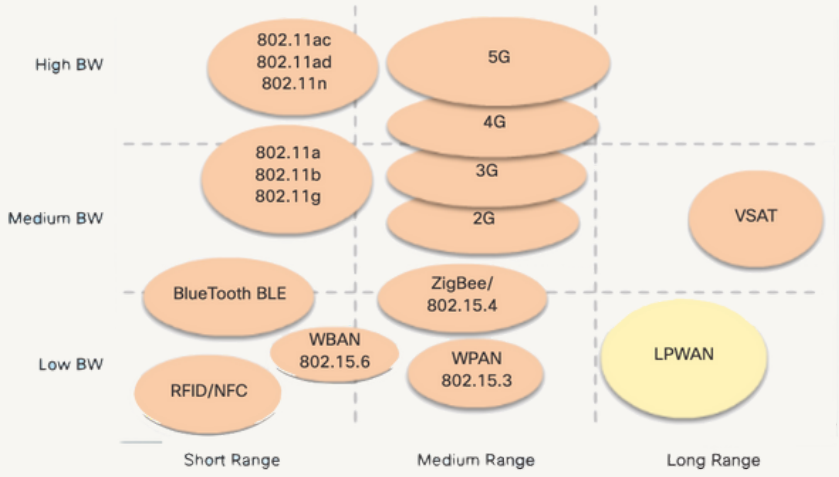
- IoT devices are integrated into all aspects of daily life.
- IoT applications carry traceable signatures and carry confidential data.
- IoT devices must adhere to a secure framework (Authentication, Authorization, Network Enforced Policy, Secure Analytics)

Technologies

WiFi

- Wireless connectivity is the biggest growth area.
- New protocols created/updated to support diverse IoT devices: ZigBee, Bluetooth, 4G/5G, LoRaWAN
- Protocols created for short, medium, and wide ranges
- Low-Power Wide-Area Networks (LPWAN) is designed to support long range communications for low bit rate devices such as sensors, actuators, and controllers

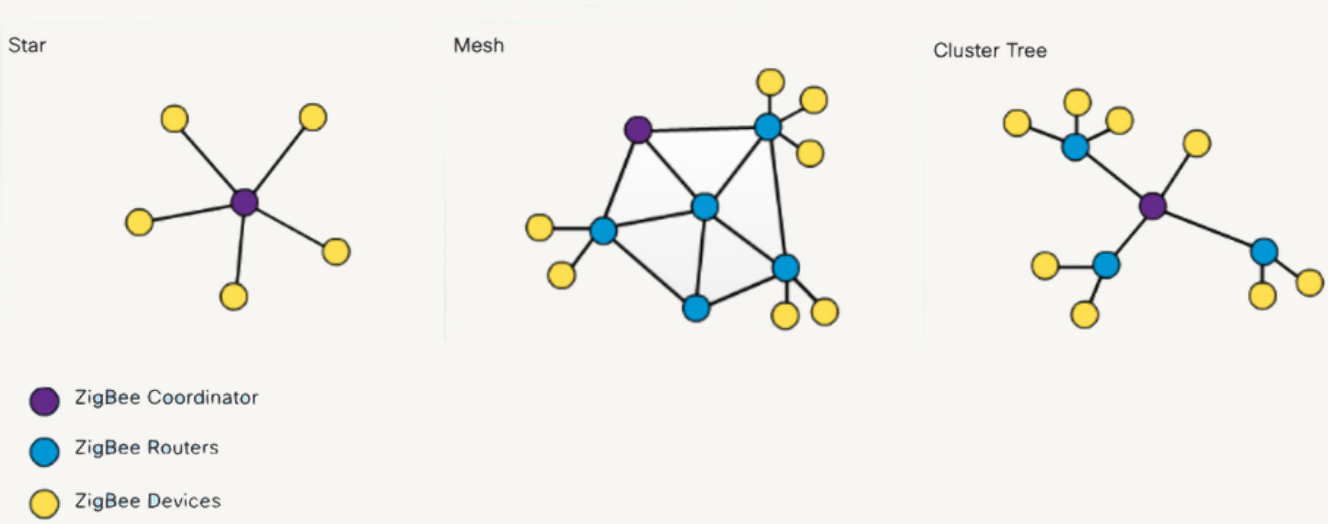
WIFI
IEEE STANDERD - WIRELESS- 802.11 this all for wifi
Difference on the speed and data transfer range



Market Name Standard	ZigBee/802.15.4	GSM	Wi-Fi/802.11	Bluetooth/802.15.1
Focus	Sensor Network/Low Power	Wide Area Voice and Data	High Speed Internet	Device Connectivity
Battery Life	years	1 week	1 week	1 week
Bandwidth	250 Kbps	Up to 2 Mbps	Up to 54 Mbps	720 Kbps
Range	Up to 100 Meters	Several Kilometers	100+ Meters	10+100 Meters
Advantages	Low Power, Low Cost	Existing Infrastructure	Speed, Ubiquity	Convenience

ZigBee

- is a low-energy, low-power, low-data rate wireless protocol
- IEEE 802.15.4-based standard specification
- max speed 250 kbps transfer rate
- Used in Home networks , Medical
- ZigBee Topologies (star , Mesh, Cluster Tree)



- intermittent data transfar - collect information from something

- ZigBee uses an Application Profile Identification Number (APID). to communicate
- Application profile ID numbers are 16-bit numbers that relate to public profiles, manufacturing profiles, or private profiles.
- ZigBee has LONG BATTERY LIFE

Bluetooth
- Bluetooth devices takes place through a short range wireless network called a personal area network (PAN).
- IEEE 802.15.1
- IOT technology : Bluetooth Low Energy (BLE) also known as Bluetooth Smart
- speed (1 Mbps) its faster then ZigBee
- Bluetooth 5 bandwidth is defined at 2 Mbps

4G/5G (cellular Technology)
- 4G - if moving the speed is 100 Mbps / not moving speed is 1 Gbps
- for IOT : NarrowBand IoT (Or NB-IoT)

LoRaWAN
- Low Range WAN(wide area network)
- It's wireless wide area network Technology which can be used by low power IOT devices
- data transfer rate range from 0.3 kbps to 50 kbps
- Used Technology: Adaptive Data Rate (ADR)
- it connects like extended star topology (star-of-stars)
  advantage : high encryption schema

WPA - WIFI protected access
WPA 2 - AES

Cloud Computing Model
- There is Hardware / Software clouds that Available over Internet but you have to pay it and you should have an internet (you can access anywhere) Method : PAY-AS-YOU-GO

Cloud computing offers services in the following areas:

- Infrastructure as a Service (IaaS) – Hardware including servers and other infrastructure components are supplied by a provider and adjusted on-demand. The provider handles system maintenance, backups and continuity planning.

- Platform as a Service (PaaS) – A provider provides the platform, servers, storage, and OSs for users to develop and launch applications

- Mobile PaaS (mPaaS) – Providers supply development capabilities for mobile application designers and developers.

- Software as a Service (SaaS) – Software, such as messaging, IoT data processing, payroll processing, gaming, and tax preparation is licensed on a subscription basis and hosted on cloud servers.

could service Providers (on-demand)

- Amazon AWS
- IFTTT : If this then that
- Zapier - business - driven
- Built.io - creating applications such as moblie apps
- Webex Teams - for IOT

Fog Computing Model

- Distributed computing infrastructure closer to the network edge.
- Edge devices run applications locally and make immediate decisions
- Reduces the data burden on networks as raw data not sent over network connections.
- Enhances security - keeping sensitive data from being transported beyond the edge where it is needed.
- Fog applications monitor or analyze real-time data from network-connected things and then take action such as locking a door, changing equipment settings, applying the brakes on a train, zooming in with a video camera,
- The action can involve machine-to-machine (M2M) communications and machine-to-people (M2P) interaction
- Cisco predicts that 40% of IoT-created data will be processed in the fog by 2018

Big Data

- Data in Motion -- Moving one to another
- Data in rest -- saving data

Big Data is typically characterized in three dimensions: volume, velocity, and variety.

3 V's

Volume - amount of data being transported and stored quantity of data (how much data is available) -(35 zettabytes)

Velocity - rate at which this data is generated (how often the data is generated)

Variety - different types of data

projects that deal with various aspects of storing, computing, and transmitting big data sets are :

- Apache Hadoop,
- Webex Teams,
- Cassandra,
- and Kafka.

Data Storage

•IoT devices may store data for a period of time before sending it out for processing. – especially for devices that do not maintain constant connections to their gateways or controllers.

•Critical that all IoT storage devices encrypt data for storage to avoid data tampering or theft

•Self-encrypting drives have encryption capability built into the drive controller - encryption and decryption done by the drive itself, independent of the operating system.

•Self-encrypting flash memory – manufacturers beginning to release new devices with self-encrypting flash memory

Data Transmission

•If data is not properly secured through encryption, it can be intercepted, captured or manipulated while in transit.

•Modern encryption algorithms may require more processing power than what is available in the IoT device.

•As well as physical security, IoT devices must be able to protect its own firmware and the data it transmits.

•Ensure that IoT devices are running the latest version of their firmware and protocols.

•Common attack: trick devices into using sub-optimal security parameters under which the connection can be exploited

•Servers, cloud endpoints, intermediary devices should also be secured and use strong encryption algorithms before communicating with IoT devices.

CH 5: THE CISCO IOT SYSTEM

Device Management
Control Plane, Data Plane, Management Plane

- Control plane is the brains of the device, used to make forwarding decisions.
- Data plane is activities done to receive data from other devices and to forward them to the next device
- Management Plane allows connection to modify a configuration or update software running on a device.

Securing the Control, Data, and Management Planes in IoT

- Securing the data plane relates to secure data as it crosses network devices.
- Securing the control plane relates to securing the network device itself with tools such as passwords and data encryption.
- Securing the management plane is secured by updating software and firmware with the latest patches

Securing the Data and Management Planes in IoT

- The management plane is usually secured by updating the operating system (OS) (or firmware)

Management Planes - secure, Updating OS / Firmware

to minimize the problem of compromising the integrity of your network by incorporating an IoT device:

- Make sure the new IoT device can be easily updated.
- Buy from a reputable manufacturer.
- Connect IoT devices to a different segment of the network or connect them to a different Virtual LAN (VLAN).
- Check for and apply system updates regularly.
- Make sure all the default usernames/passwords have been changed.
- Make sure management access to the device is only permitted from a trusted network source.
- Turn off all unnecessary services

Control plane - secure, Strong password and data encryption

Data Plane - secure,

Types of Markets

Horizontal Market

- Meet common or similar needs for a wide range of industries.
- Security, information technology, and finance companies are examples of industries that operate in horizontal markets..

Vertical Market

- Offer goods and services to a set of customers with specialized needs.
- Automotive, banking, education, healthcare, retail, and technology are considered vertical markets

Horizontal Markets	Vertical Markets
- Security - Information Technology - Finance	- Electronic Medical Records - Plant Automation - Substation Automation - Smart Traffic & parking

Challenges in Healthcare

- An increasingly aging population
- High demand services
- Shortages in key medical specialties
- Rising healthcare costs

Cisco Care-At-A-Distance Solutions

- Cisco Extended Care - Facilitates remote patient engagement and care team interactions
- Cisco TelePresence for Healthcare - Links people globally for training, consultation, and specialized collaboration
- Cisco WebEx for Healthcare - Facilitates remote collaboration, training, and patient education
- Cisco Virtual Patient Observation - Monitors patients remotely from a centralized location
- Cisco Patient Connect - Delivers personalized medical information, streaming media, and voice and video services to any device in the patient room
- Cisco Healthcare Intelligent Contact Center - Equips call center agents to proactively help patients
- Cisco Context-Aware (Location-Aware) Healthcare - Integrates location and sensor information to respond faster to patients
- Digital Media Suite for Healthcare - Centrally manages healthcare education and training

- Cisco Services for Connected Health - Leverages healthcare experts to plan, build, and manage a network that meets clinical, business, and compliance needs.
- Cisco Medical-Grade Network - Creates a network foundation that supports growth and the specific needs of healthcare.

Challenges Faced By Modern Cities

- Overcrowding
- Increasing pollution
- Increasing traffic congestion
- Inadequate parking
- Inefficient use of street lighting, water, and waste management
- Need for continued growth
- Pressure to provide safer and more secure cities
- Budget and resource constraints

Cisco Smart+Connected Solutions

- Customer segments of a city include its citizens, visitors, industry partners, businesses, and municipal operations.
- Smart cities must address the needs of these segments.
- Smart City Value Propositions:

Lighting, Operations Centers, Parking, Safety and Security, Traffic, Wi-Fi.

Cisco Smart+Connected Wi-Fi

- Citizen Services - Public Wi-Fi allows citizens to access the Internet.
- City Services - Location-based services help city planners acquire near real-time data.
- Business Services - Virtual tourist applications provide visitors with access to historical facts as well as ideas for visiting landmarks.
- City Commerce - Merchants publish targeted offers and deals to citizens based on their location.
- Infrastructure Management Services - Parking applications provide citizens with information on real-time parking availability. Traffic applications give traffic operators a real-time view of live traffic conditions and incidents, thus reducing response time.

Cisco Smart+Connected Lighting

- Drastically reduce city energy consumption, costs, and maintenance
- Improve citizen vehicle compliance and increase violation capture and city revenues
- Enhance situational awareness, real-time collaboration, and decision making across city agencies
- Add intelligent, sensor-based IoT innovations to transportation, utilities, public safety, and environmental monitoring without adding significantly more physical infrastructure

Cisco Smart+Connected Parking and Traffic

- Smart cities can simplify parking and improve traffic flow.
- The Cisco Smart+Connected Parking solution provides citizens with real-time information about available parking.
- Also allows them to book spaces in advance using mobile applications.

Cisco Smart+Connected Operations Center

- Cities are increasingly looking for a customized, integrated, single-interface view of this data.
- The Cisco Smart+Connected Operations Center solution displays sensor, map, and video data across a single layout.
- It allows operators to control dynamic activities involving image processing, video feeds, data integration, and alerts.

Challenges in Energy

- Rapid increase in consumption is putting a strain on energy providers in many countries.
- There is also an increasing pressure to use low-carbon energy sources instead of fossil fuels.
- Different ways of thinking about power and the way that it is consumed are needed.

IoT Solutions for the Power Grid

- Utilities need a more modern and agile electric grid.
- Smart grid provides more complex interconnections between the producers, storage facilities, and consumers of electricity.
- Smart grid brings the notion of the consumers generating power for themselves and to the grid.

Cisco Smart Grid Solutions

•Cisco provides many smart grid solutions including:

- oGridBlocks Architecture
- oConnected Grid Services
- oField Area Network
- oTransmission and Substation
- oGrid Security
- oGrid Operations

Challenges in Manufacturing

- Manufacturing must continually integrate new innovative technology into the existing plant infrastructure.
- Multiple siloed operational technology networks become a problem.
- Diversity in networks increases cost and complexity.
- That lack of integration leads to a broad range of issues, including:
    - Inefficient operations
    - Slow response times both in the factory and in the market
    - Poor quality control
    - High overhead
    - Compromised security

Cisco provides the following IoT manufacturing value propositions:

- Cisco Connected Factory - Provides improved security, operational efficiency and productivity using a unified, factory-wide, secure network

- Cisco Connected Machines - Provides business improvements, such as overall equipment effectiveness (OEE), machine monitoring, machine and process optimization, and predictive maintenance

- Cisco Secure Ops - Provides protections against risks, improves efficiency, and reduces site downtime

- Cisco Connected Supply Chain - Provides increased supply chain flexibility and agility

- Cisco Communications and Collaboration Tools - Improves communication and reduces downtime

CH 6:

The Sustainable Development Goals

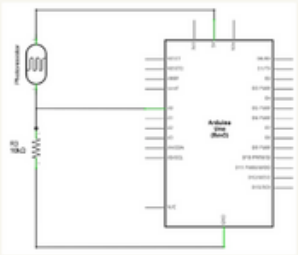
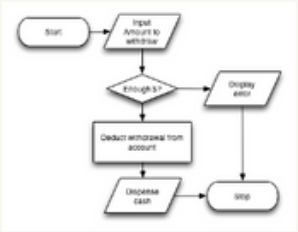
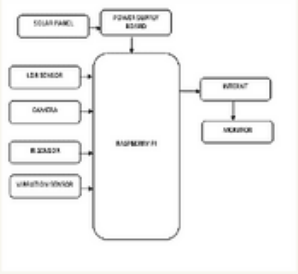
189 countries trying to solve problems - MDG - first then SDG

In 2015, 189 world leaders at the United Nations Sustainable Development goal Summit unanimously adopted the 2030 Agenda for Sustainable Development.

The Engineering Design Process
There is 5 steps :

- Inspire / Empathize
- Define / Ideate
- Prototype
- Test/Refine
- Present / Communicate

Types of Diagrams

The circuit layout Schematics	An important next step. Electronic components have very specific requirements such as power, polarity and connections.	
Flowcharts	Used to represent processes or workflows Step by step	
Sequence diagrams	Also called event diagram or event scenario, is used to represent interactions between entities along a timeline	
Block diagram		

after Drawing Diagrams , Do the prototype then do the coding then test then present in the market and watch if there is a problem

Business Model Canvas - 9 blocks

1. Customer Segments
2. Value Proposition
3. Channels
4. Customer Relationship
5. Revenue Stream
6. Key Resources
7. Key Activities
8. Key Partnerships
9. Cost Structure

Customer Interface Management

1. Value Proposition : Identifies the value to the customers
2. Customer segment : Identify the specific people who will be most attracted to the value proposition.
3. Channels : how to deliver the product to the customer
4. Customer Relationships : call center , customer services

Infrastructure Management

1. Key Activities : Identifies which activities the organization is required to perform well.
2. Key Partnerships : Identifies who can help the organization leverage the business model.
3. Key Resources : Identifies the infrastructure required to create value and shows which assets are indispensable in the organization model.

Business Finance

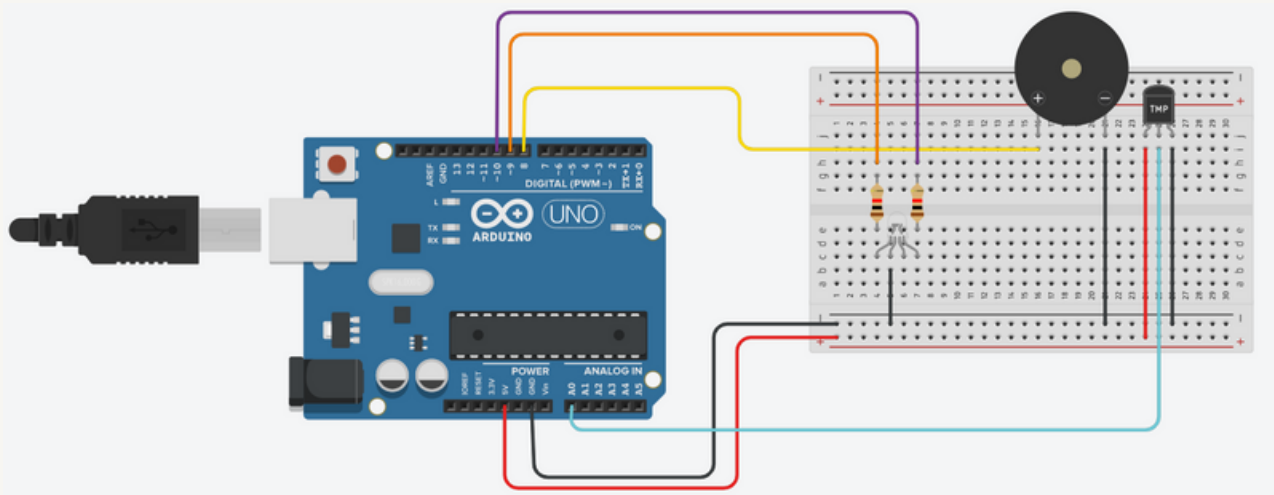
1. Cost : Identifies the cost of the operation.
2. Revenue : Identifies how and through which pricing mechanism the organization is capturing value.

Example : Restaurants ----

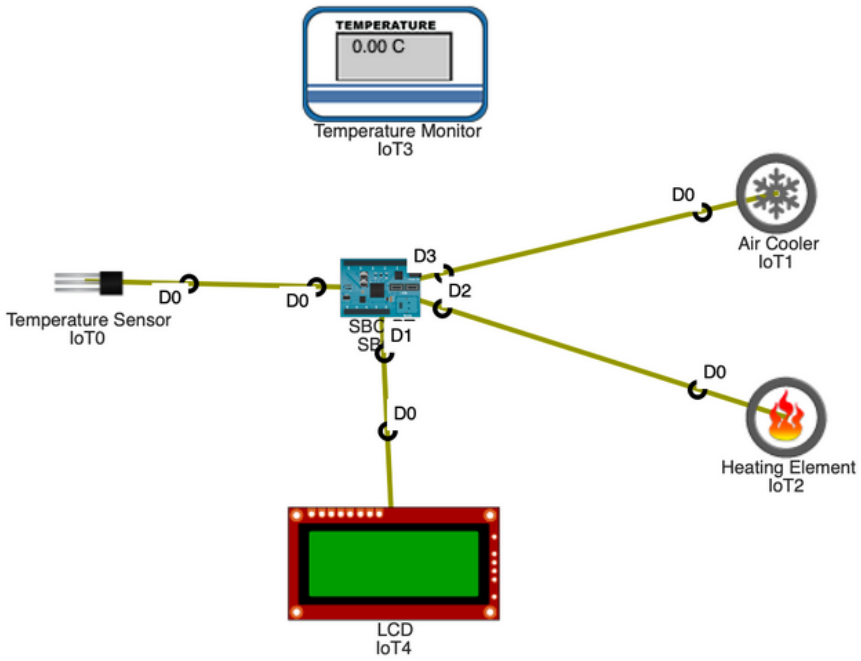
- Value Proposition : quality food / services , tasty, discount
- Customer segment : any one can come who has money and want food
- Channels : Waiter, apps, websites to order
- Customer Relationships : call center

- Key Activities : sell food , collect order, deliver food
- Key Partnerships : farmer for vegetables , driver
- Key Resources : important resources like chief

- Cost : buy material, rent for showroom, pay salary
- Revenue : selling food item




```
int sensePin = A0; //This is the Arduino Pin that will read the sensor output
int sensorInput; //The variable we will use to store the sensor input
double temp; //The variable we will use to store temperature in degrees.
int buzzer = 8;
int red = 9;
int green = 10;

void setup() {
// put your setup code here, to run once:
Serial.begin(9600); //Start the Serial Port at 9600 baud (default)
}

void loop() {
// put your main code here, to run repeatedly:
sensorInput = analogRead(A0); //read the analog sensor and store it
  //till here only on need for calculation
temp = (double)sensorInput / 1024; //find percentage of input reading
temp = temp * 5; //multiply by 5V to get voltage
temp = temp - 0.5; //Subtract the offset
temp = temp * 100; //Convert to degrees

Serial.print("Current Temperature: ");
Serial.println(temp);

if(temp>50)
{
digitalWrite(buzzer, HIGH);
digitalWrite(red, HIGH);
digitalWrite(green, LOW);
}
else
{
digitalWrite(buzzer, LOW);
digitalWrite(green, HIGH);
digitalWrite(red, LOW);
}
}
```




```python
from gpio import *
from time import *

def main():
    pinMode(0, IN)
    pinMode(1, OUT)
    pinMode(2, OUT)
    pinMode(3, OUT)
    print("Blinking")

    while True:
        temp= digitalRead(0);
        print("Temprature", temp);

        if(temp>520):
            digitalWrite(3, HIGH);
            digitalWrite(2, LOW);
            customWrite(1, "cooler-on");
        else:
            digitalWrite(3, LOW);
            digitalWrite(2, HIGH);
            customWrite(1, "Heater-on");
        delay(1000);

if __name__ == "__main__":
    main()
```